Mohamed Tarek Abdelhady

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering fresh graduate with a strong foundation in Embedded Systems, Industrial Automation, PCB Design, Electronics, IoT, and Renewable Energy. Skilled in PLC programming (Siemens TIA Portal), classic control systems, embedded C, PCB design, and hardware–software integration through academic projects and hands-on training. Passionate about automation and control systems, with a commitment to continuous learning and solving real-world engineering challenges. Active IEEE and GDSC member seeking a Junior Automation Engineer role to apply technical knowledge and contribute to innovative automation solutions.

EDUCATION

Bachelor of Engineering in Electronics & Communication Engineering Graduated Jul 2026
Higher Institute of Engineering and Technology in Manzala, Egypt

- **Grade:** Good

GRADUATION PROJECT

Design, Modeling, Implementation, and IoT-Based Monitoring of a 60 W photovoltaic Renewable Energy System (Graduation Project, Grade: Excellent)

- Designed and simulated a 60W IoT-monitored photovoltaic system using MATLAB/Simulink.
- Implemented a TL494-based buck converter for efficient battery charging regulation.
- Developed a conventional inverter using 555 timer IC for basic AC power conversion.
- Implemented an EGS002-based SPWM inverter and evaluated output voltage waveform, frequency stability, and conversion performance under different load conditions.
- Created IoT monitoring board for real-time data collection and analysis using Firebase.
- Developed a Firebase-based web dashboard for real-time monitoring of the system.

TECHNICAL SKILLS

- **PCB Design & Electronics:** Altium Designer, Schematic Capture, PCB Layout, Component Placement, Routing, DRC, Digital Electronics, Power Electronics, Electronic Circuit Design
- **Embedded Systems & IoT:** Arduino, Embedded C, IoT Systems, Hardware–Software Integration
- **Automation & Control:** PLC (Siemens TIA Portal, Ladder Logic), Classic Control, VFD, Control Systems, Industrial Panel Building
- **Software & Tools:** Proteus, Multisim, Factory I/O, CADe_SIMU, EKTS, Arduino IDE
- **Programming:** Python, Embedded C, Ladder Logic

PROFESSIONAL EXPERIENCE

HA (Hassan Ali Consulting Group) Aug 2025
Industrial Automation Trainee 140 hours

- Completed 140 hours of practical training in Industrial Automation, covering PLC Programming, Classic Control, Motor Drives, and Industrial Panel Building.
- Developed and debugged 20+ PLC programs using Siemens TIA Portal, validated automation sequences through Factory I/O, and configured VFDs for three motor control applications.

Petrobel (Belayim Petroleum Company) Jul 2025
Engineering Intern 56 hours

- Observed manufacturing operations and production workflows, gaining exposure to industrial equipment, process monitoring, and quality control procedures.
- Assisted engineering teams in understanding production processes and operational efficiency.

3M Company Aug 2024
Engineering Intern 160 hours

- Supported manufacturing process optimization, production monitoring, and operational improvements.

Self-Study — PCB Design *Altium Designer* Self-Learning

- Practiced schematic capture, footprint assignment, PCB layout, and Design Rule Checking (DRC) using Altium Designer.

PROJECTS

PLC Programming Projects (Siemens TIA Portal)

- Developed and tested PLC programs for industrial applications: Color Detection, Pick-and-Place, Elevator Control, Automatic Filling Tank, Traffic Light Control, Cyclic Motor Operation, and Temperature Monitoring.
- Developed and tested Siemens PLC Ladder Logic programs using timers, counters, comparators, and arithmetic instructions to automate industrial sequences.

Classic Control Projects

- Designed and implemented control circuits: Sequential Motor Control, Star-Delta Starter with Forward/Reverse, Automatic Washing Machine Controller, and Timed Motor Operations.
- Created detailed wiring diagrams and tested circuit functionality under various load conditions.

Smart Blind Glasses

- Developed a wearable assistive device using ultrasonic sensors for obstacle detection and navigation assistance.
- Implemented dual haptic (vibration motor) and auditory (buzzer) feedback, and designed a portable, low-power solution refined through user testing.

RFID-Based Attendance & Person Counter

- Designed an RFID-based system for automated attendance tracking and room occupancy counting.
- Implemented occupancy-based automated lighting control for energy efficiency and built an administrator interface for data visualization and access.

CERTIFICATIONS & TRAINING

- **Industrial Automation Training** 140 hours, Aug 2025
  Classic Control, PLC Programming, Motor Drives, Panel Building — HA Consulting Group
- **CCNA Training** 24 hours, Jul 2024
  Networking fundamentals, routing, switching, and network security
- **Python & Machine Learning** 24 hours, Jun 2025
  Python programming, data analysis, and machine learning fundamentals
- **Home Automation Training** Jul–Aug 2024
  Smart home systems, IoT integration, and automation protocols
- **Network Fundamentals L1** Jul 2023
  Basic networking concepts and protocols

LEADERSHIP & ACTIVITIES

- **IEEE Member (Vice Chair 2025)** — Led student chapter activities, organized technical seminars, and mentored junior members; active in technical workshops and engineering community events.
- **GDSC Technical Head (2025)** — Engaged in collaborative technology learning, development initiatives, and hackathons; contributed to open-source projects and community development.

MILITARY STATUS & LANGUAGES

- **Military Status:** Exempted
- **Languages:** Arabic (Native), English (Intermediate)